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AI for Bharat Hackathon

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Team Name : WIN.EXE

Team Leader Name : Khushi Agarwal

Problem Statement :

Brief about the Idea:

ALMOND - Your Team's Memory for Bug Fixes

The Problem We're Solving:

Engineers leave companies every ~2 years, taking their knowledge with them
Teams waste hours fixing bugs that were already solved months ago
The solution is buried somewhere in old Slack threads or closed PRs

What ALMOND Does:

Think of it as your team's external brain. Instead of asking "Has anyone seen this error before?", ALMOND searches through your entire deployment history and tells you "Yes, Sarah fixed this 6 months ago in PR #402. Here's what she did."

In Simple Terms:

ALMOND remembers everything your team has ever fixed, so you don't have to solve the same problem twice.

What Makes Us Different

The Old Way (RAG/Vector Search):

Looks for "similar" errors like a keyword search

Finds "Error 500" but doesn't understand WHY it happened

Misses the story: "This broke because of that database change last week"

The ALMOND Way:

Actually reads through your history.

Connects the dots: "Error started → after PR #402 → which changed the database"

Gives you the exact fix that worked before, not generic advice

Why Teams Love It:

No more fixing the same bug on different teams

Junior devs get instant answers without bothering seniors

Only suggests fixes your team has already used successfully

No random AI suggestions that might break things

Fast & Cheap:Answers in 2 seconds instead of 45

Runs real code to verify everything before telling you

What It Actually Does

The Three-Tier Brain:

Tier 1: Your Team's History (The Good Stuff)

Searches actual PRs and fixes your team made

Tier 2: The Smart Cache

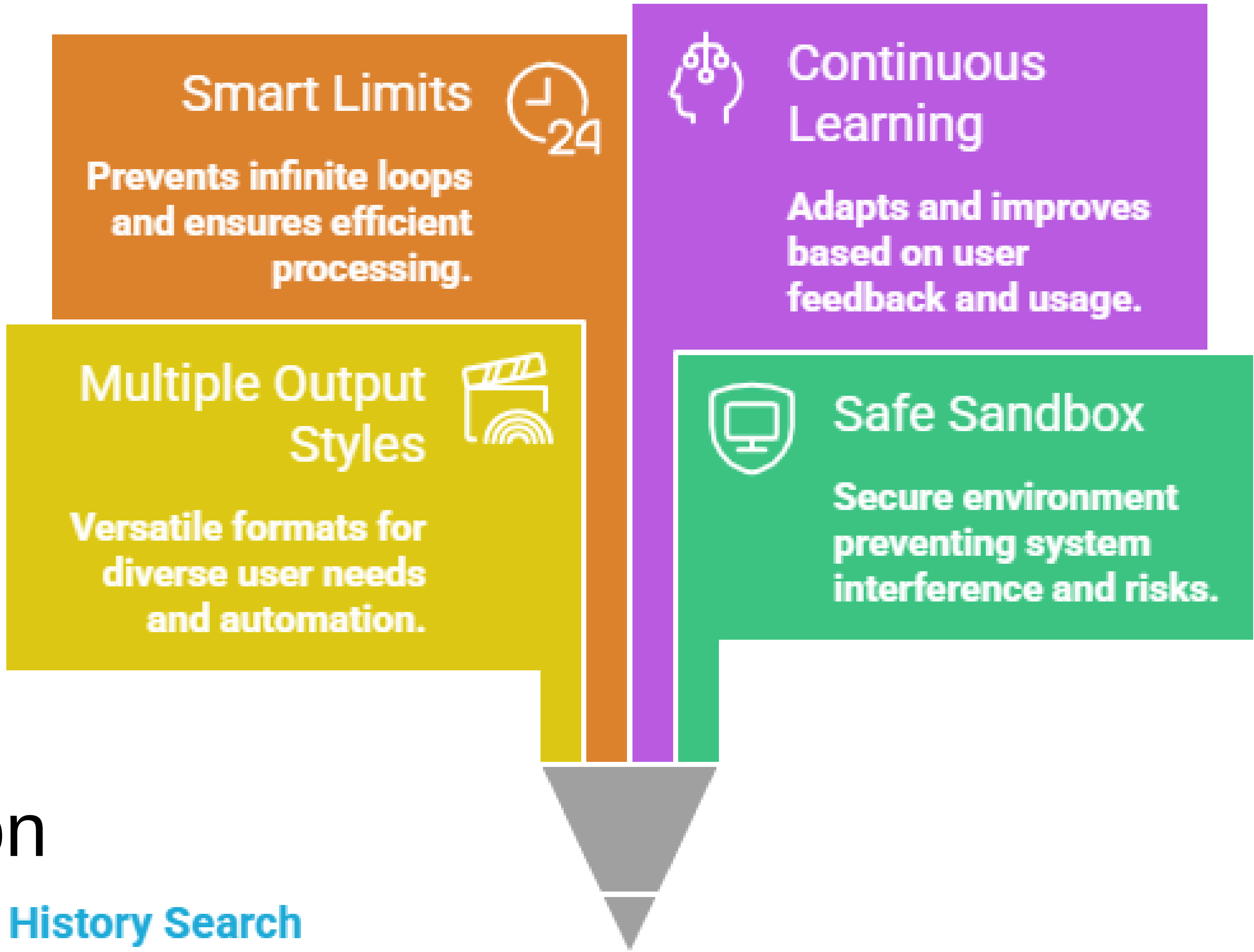
Remembers AI-suggested fixes that worked

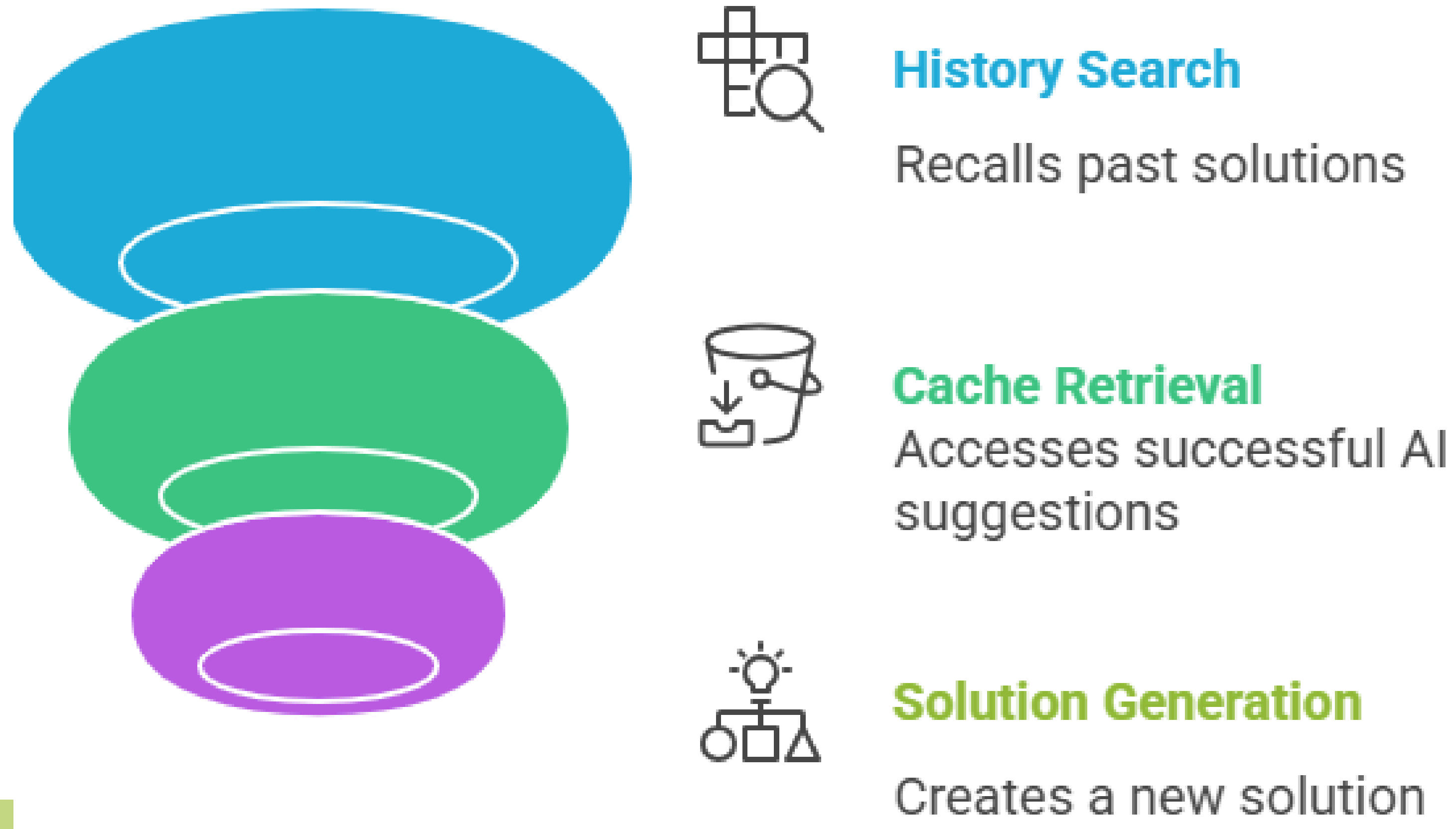
Gets smarter every time someone accepts a fix

Tier 3: Figure It Out Mode

When it's a totally new problem, generates a fresh solution

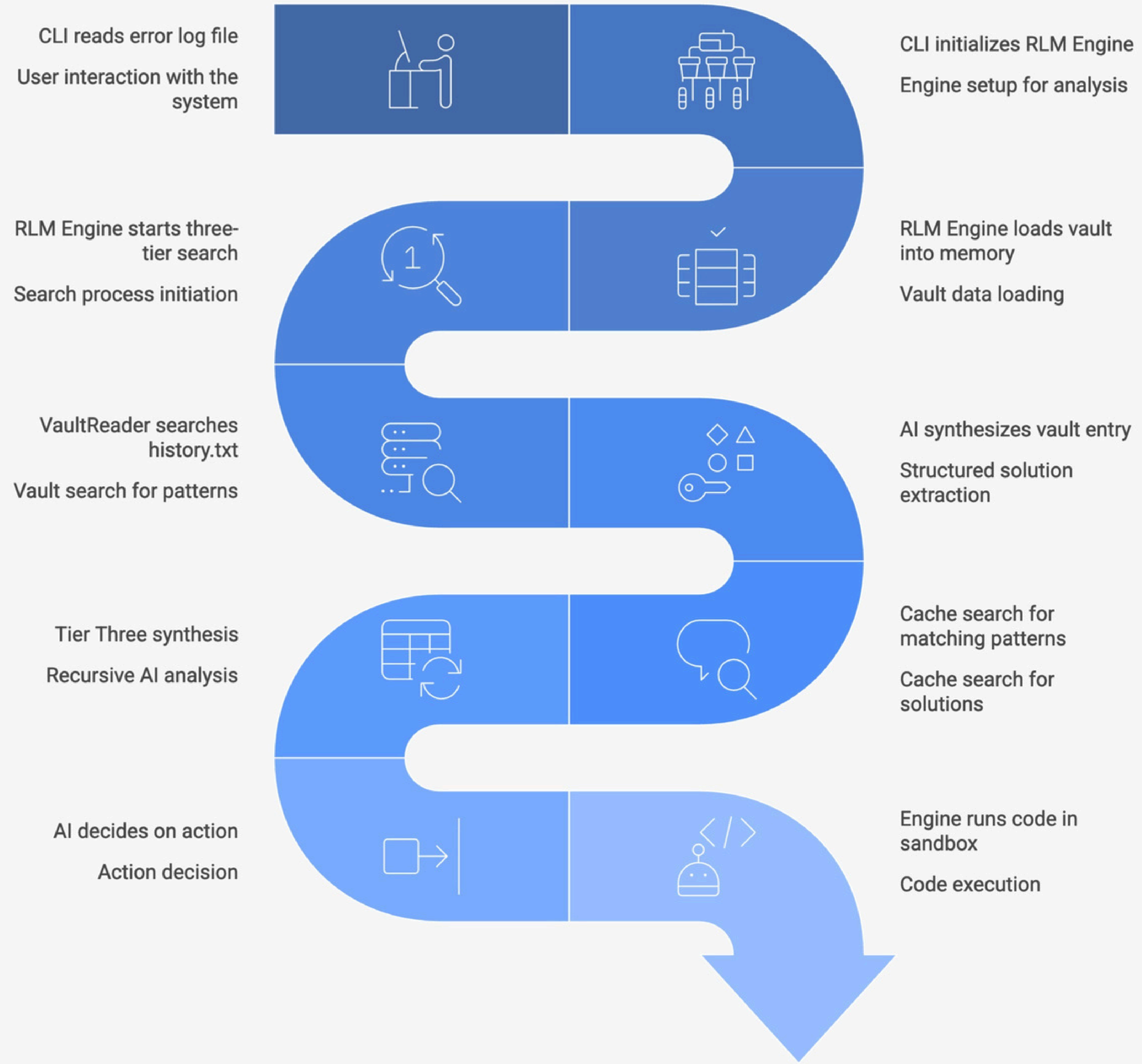
Then saves it for next time



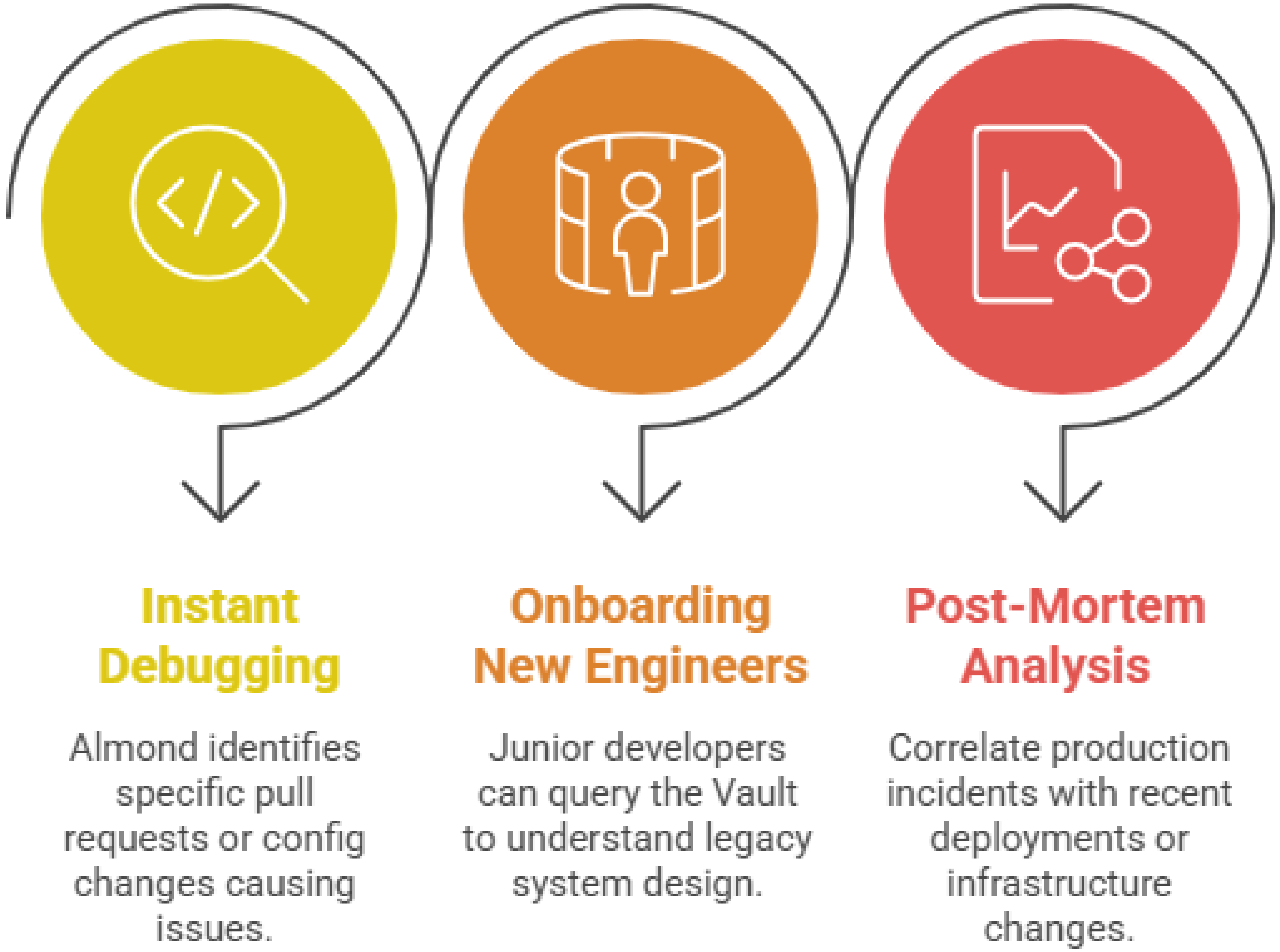


Process flow diagram or Use-case diagram

ALMOND Data Flow for Error Debugging



Almond Use Cases



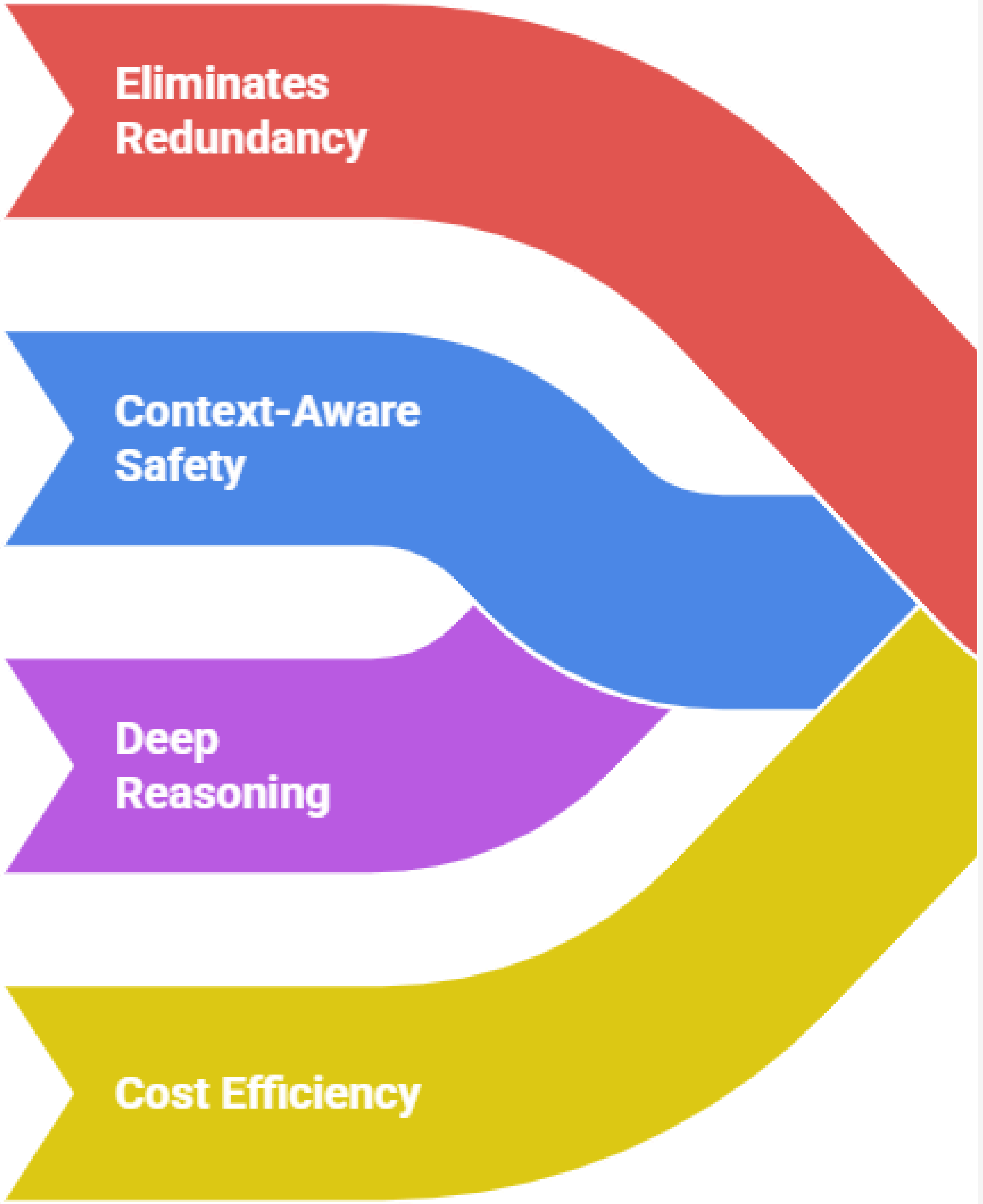
Wireframes/Mock diagrams of the proposed solution

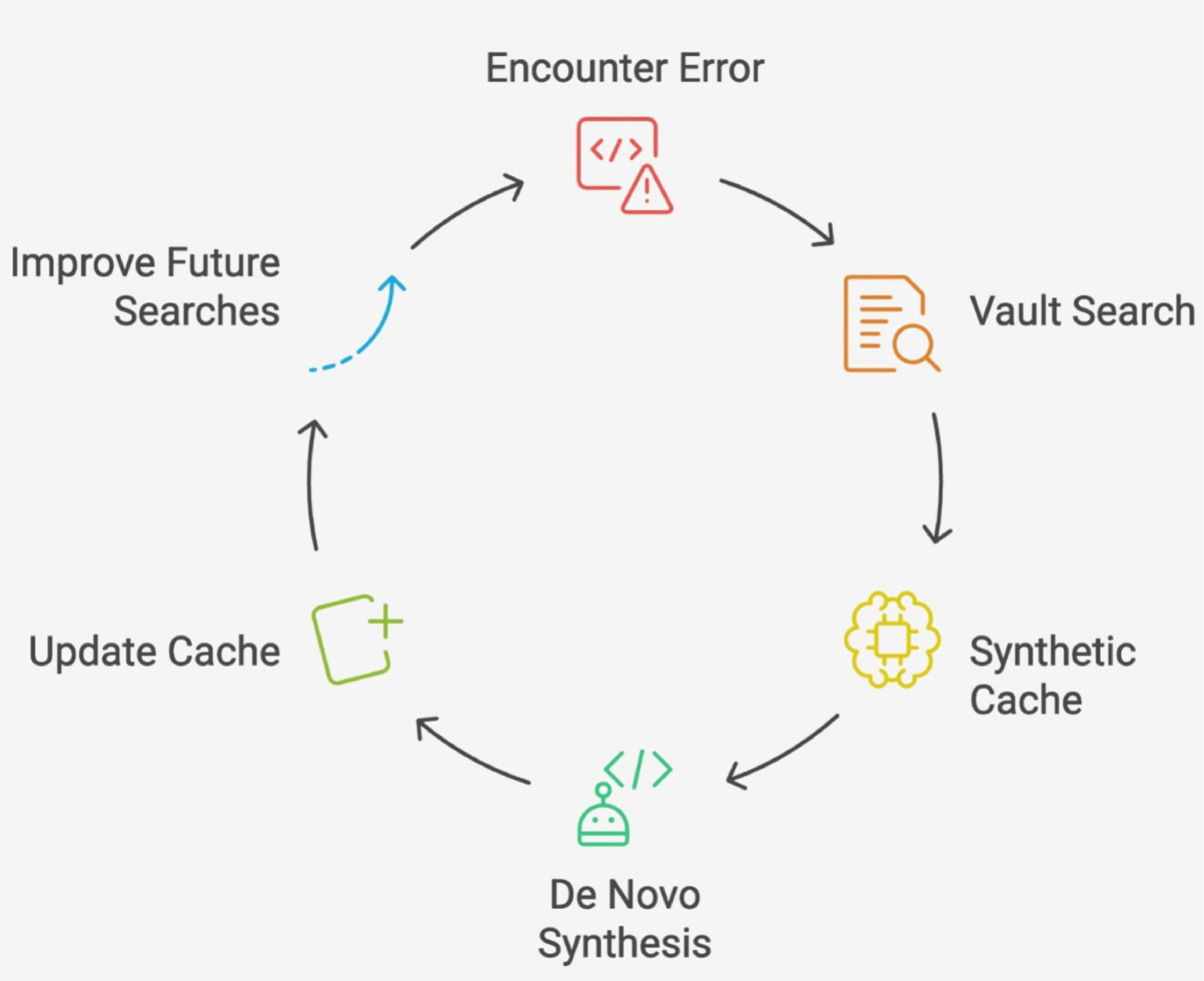
1 Prevents repetitive bug fixes across different teams.

2 Ensures secure coding suggestions based on team history.

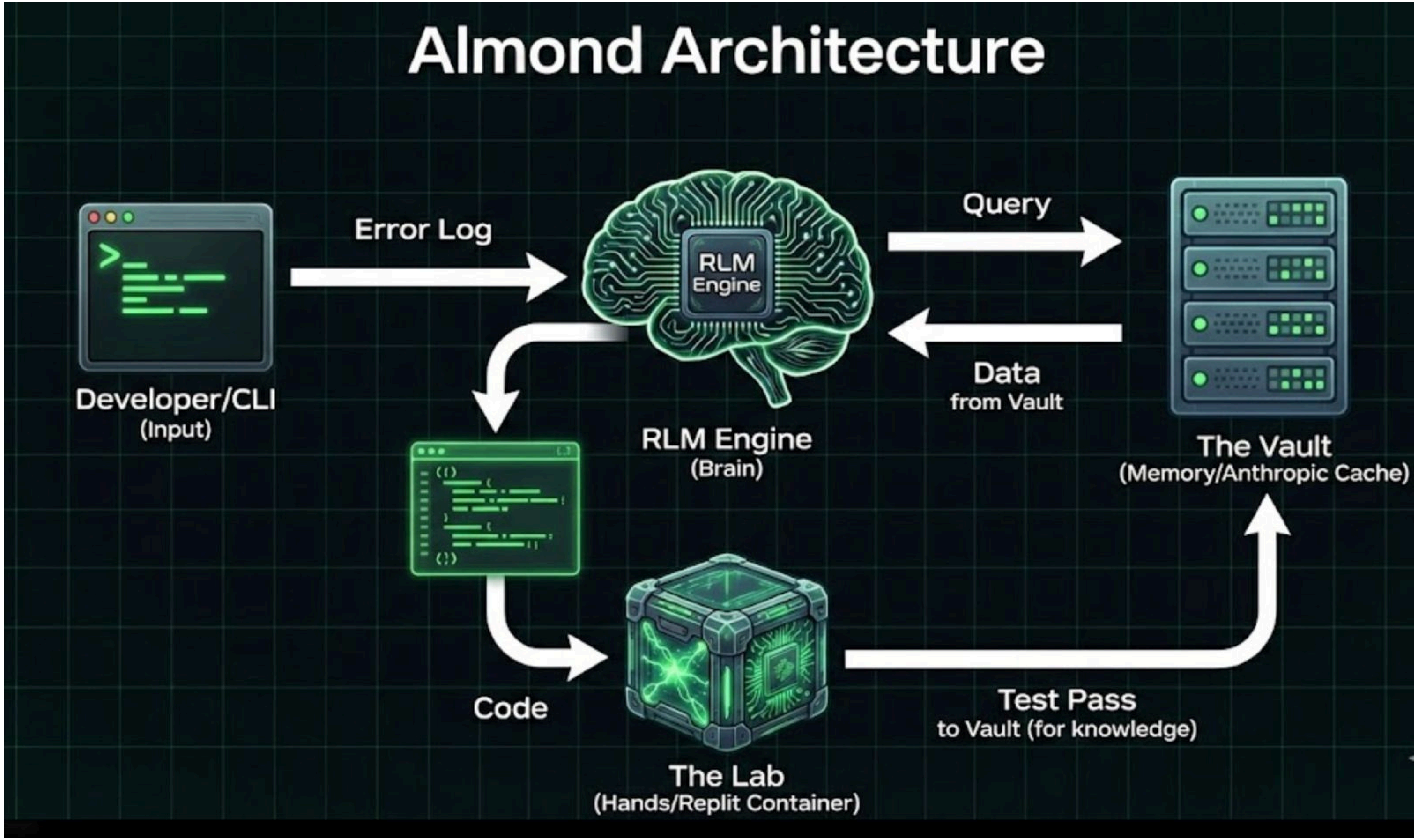
3 Understands complex causal relationships in code.

4 Optimizes resource usage through context caching.





Architecture diagram of the proposed solution:



Technologies to be used in the solution:

Core Logic: Recursive Language Model (RLM) based on Zhang et al.

LLM Backend: Anthropic Claude 3.5 Sonnet (via Context Caching API).

Environment: Sandboxed Python REPL for safe recursive code execution.

Interface: Python-based CLI tool.

Estimated implementation cost (optional):

Development Cost:

• $0 - Completely open source

Monthly Operating Costs:

Small Team (10-50 queries/day): • $15-20/month • Less than a daily coffee

Medium Team (200-500 queries/day): • $60-100/month• Less than a weekly team lunch

Large Team (1000+ queries/day): • $300-500/month • Less than 1 engineer's hourly rate

Return on Investment:

• Saves 5 hours/week per developer

• 10-person team saves $10,000/month in time

• Costs only $60/month

• 166x ROI

Add as per the requirements for the hackathon:

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Thank You